

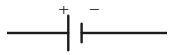
Recitation 1 - Basic Circuits

0 The one rule that prevents most Week-1 mistakes

Every quantity in this course describes a different *kind* of thing. Get this straight now and half the confusion disappears.

Quantity	Symbol	Unit	Describes	How you measure it
Voltage	V	volt (V)	a difference across two points	meter in parallel , very high internal resistance
Current	I	ampere (A)	flow through one path	meter in series , very low internal resistance
Resistance	R	ohm (Ω)	a property of the component itself	measure the part on its own, with power off

1 Charge, current, and voltage



Charge Q , measured in coulombs (C). One electron carries $q_e = -1.602 \times 10^{-19}$ C, so 1 C is the charge of about 6.24×10^{18} electrons. In the symbol, the **long plate is the positive terminal** and the short plate is negative. That convention appears on every schematic in this course.

Battery / DC source

Current is the rate at which charge passes a point:

$$I = \frac{Q}{t} \quad 1 \text{ A} = 1 \text{ C/s}$$

Voltage is energy delivered per unit charge:

$$V = \frac{E}{Q} \quad 1 \text{ V} = 1 \text{ J/C}$$

A 9 V battery hands 9 joules of energy to every coulomb of charge that passes through it. In a lamp, that energy leaves as light and heat.

Two directions, one convention. Electrons physically drift from the $-$ terminal toward the $+$. *Conventional current*, which is what every formula and every circuit diagram in this course uses, points from $+$ to $-$. Use conventional current for all analysis; the electron direction only matters when you are explaining the physics inside the material.

Voltage can exist with no current. An open circuit (switch off, or a disconnected battery on the shelf) has full voltage across the gap and $I = 0$. Current requires a complete loop.

Problem 1

A current of 1.5 A flows through a wire for 3.0 seconds.

- (a) How much charge passes through?
- (b) How many electrons is that?

2 Ohm's law

$$V = I \times R \quad I = \frac{V}{R} \quad R = \frac{V}{I}$$

Two proportionalities to internalize:

- Current is **directly proportional** to voltage. Double V, double I.
- Current is **inversely proportional** to resistance. Double R, half I.

Problem 2

A 220Ω resistor is connected across a 5.0 V supply.

- (a) Find the current, in mA.
- (b) Find the power dissipated, in mW.

Problem 3

A lamp has a resistance of 240Ω .

- (a) Plugged into a US outlet at 120 V, what is the current and the power?
- (b) Plugged into a UK outlet at 230 V, what is the current and the power?
- (c) The voltage not quite doubled. Why did the power nearly quadruple?

3 Resistance and resistors



Resistor

$$R = \rho \frac{L}{A}$$

where ρ is resistivity ($\Omega \cdot \text{m}$), an intrinsic property of the material, L is length (m), and A is cross-sectional area (m^2).

- Longer wire $\rightarrow$ more resistance.
- Thicker wire $\rightarrow$ less resistance. **Careful:** doubling the *diameter* quadruples the area, so it cuts the resistance to one quarter, not one half.
- Different materials have very different ρ . Temperature changes it too.

4 Power, energy, and ratings

Power is energy per second. Starting from $P = E/t$ and $E = VQ$:

$$P = \frac{VQ}{t} = V \times I$$

and substituting Ohm's law gives the three forms you will use constantly:

$$P = VI = I^2R = \frac{V^2}{R}$$

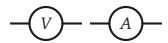
Energy over a time interval: $W = P \times t$, in joules. (Your utility bill uses kilowatt-hours, which is the same idea with bigger units.)

Problem 4

You have a 100Ω resistor rated at $\frac{1}{4} \text{ W}$.

- What is the maximum voltage you may apply across it? What current does that correspond to?
- A classmate connects it directly across a 9 V battery. How much power does it now dissipate, and by what factor is the rating exceeded?

5 Measuring: voltmeter and ammeter



Voltmeter
Ammeter

Voltmeter — connect **in parallel**, across the two points of interest. Its internal resistance is very high (about $10 \text{ M}\Omega$) precisely so that almost no current is diverted into the meter and the circuit is barely disturbed.

Ammeter — connect **in series**, in the path whose current you want. Its internal resistance is nearly zero so it does not noticeably reduce the current it is trying to measure.

Problem 5

A 6.0 V battery drives a single 100Ω resistor. Assume the battery has an internal resistance of 0.5Ω .

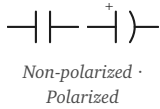
- Connected correctly: what does an ammeter in series read, and what does a voltmeter across the resistor read?
- A voltmeter ($R = 10 \text{ M}\Omega$) is mistakenly connected *in series* with the resistor. What is the circuit current now, and what does the voltmeter display? Why is that reading so misleading?
- An ammeter ($R \approx 0$) is mistakenly connected *in parallel*, straight across the battery terminals. What current flows? What happens to the meter?

Part (c) is a **short circuit**: a near-zero-resistance path that lets current run away, since $I = V/R$ with $R \rightarrow 0$. This is why multimeters have a separate jack and an internal fuse for the current ranges.

6 Batteries in series and parallel

- **Series** (end to end): voltages add, respecting polarity. $1.5 \text{ V} + 1.5 \text{ V} = 3 \text{ V}$. Capacity (mAh) stays the same.
- **Parallel**: voltage stays the same, capacity adds. Only do this with batteries of the same nominal voltage, chemistry, and state of charge.

7 Components



Capacitor

Stores equal and opposite charge, $+Q$ and $-Q$, on two plates:

$$Q = CV$$

C is capacitance in farads (F); real boards use μF , nF, and pF. Separating the charge takes work, so the capacitor stores energy in the electric field between the plates:

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$$

The polarized type (electrolytic) has a dedicated $+$ and $-$ terminal and **fails if connected backward**; the non-polarized type (ceramic) can go either way.

Problem 6

A $100\ \mu\text{F}$ capacitor is charged to $12\ \text{V}$.

- (a) How much charge is stored?
- (b) How much energy is stored?
- (c) A power supply contains a $1000\ \mu\text{F}$ capacitor charged to $400\ \text{V}$. How much energy is that, and why is this the one on the safety poster?



Inductor

A coil of wire that stores energy in a **magnetic field**, with strength set by the current through it:

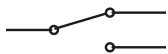
$$E = \frac{1}{2}LI^2$$

Inductance L is measured in henries (H). An inductor **resists changes in current**: change the current, the magnetic field changes, and the changing field induces a voltage that opposes the change. Common in filters and power supplies.



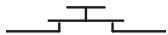
SPST switch

Single pole, single throw — two terminals, plain on/off, makes or breaks one path.



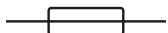
SPDT switch

Single pole, double throw — three terminals, routes the current to exactly one of two paths.



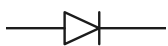
Momentary pushbutton

Closed only while held.



Fuse

A deliberate weak link: a thin element that melts when current exceeds its rating, breaking the circuit before anything more expensive is damaged. A fuse **must be in series** with the circuit it protects. A $10\ \text{A}$ fuse passes up to $10\ \text{A}$ safely and opens above that.



Diode

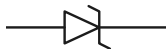
Allows current in one direction and blocks the other. Roughly: a small resistance forward, a very large resistance in reverse, with a small forward voltage drop. The triangle points in the direction of conventional current flow, and the bar is the blocking side.

What they are for: **rectification** (blocking or flipping the backward half of an AC waveform to produce DC) and **protection** (stopping reverse current and blocking spikes).



LED

A diode engineered to release its energy as visible light rather than heat. Same structure, same one-way rule, very efficient — the two outward arrows are the only difference in the symbol.



Zener diode

Deliberately designed to conduct in reverse once a specific breakdown voltage is reached; the bent ends of the bar are how you tell it apart on a schematic. Ordinary diode: rectifies current. Zener: **regulates voltage**, holding an output steady.

Problem 7

You want to drive a red LED from a $5.0\ \text{V}$ supply. The LED has a forward voltage drop of $2.0\ \text{V}$ and you want about $15\ \text{mA}$ through it.

- (a) How much voltage must the resistor drop?
- (b) What resistance do you need?
- (c) The nearest standard value is $220\ \Omega$. What is the actual current?
- (d) How much power does the resistor dissipate? Is a $\frac{1}{4}\ \text{W}$ resistor adequate?
- (e) What happens if you connect the LED with no resistor at all?

Solutions

Problem 1.

- (a) $Q = I t = 1.5 \times 3.0 = \mathbf{4.5\ C}$
(b) $n = Q / q_e = 4.5 / 1.602 \times 10^{-19} \approx \mathbf{2.8 \times 10^{19}\ electrons}$

Problem 2.

- (a) $I = V/R = 5.0 / 220 = 0.0227\ A = \mathbf{22.7\ mA}$
(b) $P = V I = 5.0 \times 0.0227 = 0.114\ W = \mathbf{114\ mW}$

Problem 3.

- (a) $I = 120 / 240 = \mathbf{0.50\ A}$; $P = 120 \times 0.50 = \mathbf{60\ W}$
(b) $I = 230 / 240 = \mathbf{0.958\ A}$; $P = 230 \times 0.958 \approx \mathbf{220\ W}$
(c) Because R is fixed, $P = V^2/R$, so power scales with the **square** of voltage: $(230/120)^2 \approx 3.7$. This is why a 120 V appliance plugged into a 230 V outlet does not merely run "a bit hot" — it fails.

Problem 4.

- (a) From $P = V^2/R$: $V_{\max} = \sqrt{P R} = \sqrt{(0.25 \times 100)} = \mathbf{5.0\ V}$, giving $I = 5.0/100 = \mathbf{50\ mA}$.
(b) $P = V^2/R = 81/100 = \mathbf{0.81\ W}$, which is **3.2× the ¼ W rating**. The resistor overheats, discolors, and fails. Nothing about Ohm's law forbids this — only the power rating does.

Problem 5.

- (a) $I = 6.0 / (100 + 0.5) \approx \mathbf{59.7\ mA} \approx \mathbf{60\ mA}$; the voltmeter reads $\approx \mathbf{6.0\ V}$ across the resistor.
(b) Total $R \approx 10\ M\Omega$, so $I = 6.0 / 10\ 000\ 100 \approx \mathbf{0.6\ \mu A}$. The resistor now drops only $I \times 100 \approx 60\ \mu V$, so essentially the entire 6 V appears across the meter and it displays $\approx \mathbf{6\ V}$. Misleading because that number looks plausible, but it is the *supply* voltage, not the voltage across the component you meant to measure — and the circuit has effectively stopped working.
(c) $I = 6.0 / 0.5 = \mathbf{12\ A}$, limited only by the source and wiring. The fuse inside the meter blows; without a fuse the meter, the battery, or both are damaged.

Problem 6.

- (a) $Q = C V = 100 \times 10^{-6} \times 12 = 1.2 \times 10^{-3}\ C = \mathbf{1.2\ mC}$
(b) $U = \frac{1}{2} C V^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 144 = \mathbf{7.2\ mJ}$
(c) $U = \frac{1}{2} \times 1000 \times 10^{-6} \times 400^2 = \mathbf{80\ J}$. Stored at 400 V and still there long after the device is unplugged — enough to injure or kill. This is why supplies carry discharge warnings and bleeder resistors.

Problem 7.

- (a) $5.0 - 2.0 = \mathbf{3.0\ V}$ across the resistor.
(b) $R = 3.0 / 0.015 = \mathbf{200\ \Omega}$
(c) $I = 3.0 / 220 = 0.0136\ A = \mathbf{13.6\ mA}$ — close enough; standard values rarely land exactly on target, and LEDs are not fussy.
(d) $P = I^2 R = (0.0136)^2 \times 220 = \mathbf{41\ mW}$, comfortably under ¼ W. Yes, adequate.
(e) The LED's own forward resistance is very small, so the current is limited only by the supply and wiring — far above the LED's rating. It burns out, usually instantly.

Vivado — Simulating Your First Design

One two-input AND gate, from a new project to reading the waveform

Everything below is simulation only. Constraints, synthesis and programming the board come later; today the goal is to check that your logic is right.

1 New project

1. **Create Project** → Next
2. Project name `lab1_and2`, location `C:\fpga`, tick *Create project subdirectory*
3. Project Type → **RTL Project**, tick *Do not specify sources at this time*
4. Default Part → **Parts** tab → search `xc7a35tcpg236-1` → Finish

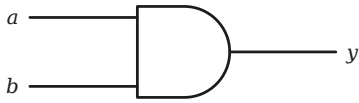
The device you select does not affect simulation at all. It only matters once you target a real board.

2 Design file

Alt+A → Add or create design sources → **Create File**.

Set File type to SystemVerilog. The dialog defaults to Verilog. If you leave it, every `logic` and `always_comb` throws a syntax error, and the message only says "syntax error" — it never mentions the file type. This is the most common thing people get stuck on in the first session.

Name the file `and2`. When the **Define Module** dialog appears, click OK and then Yes — typing the ports in code is faster than filling in that form.



this AND gate

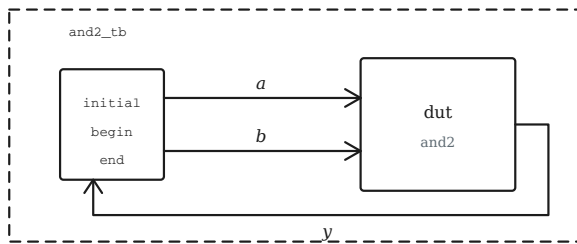
```

module and2(
    input logic a,
    input logic b,
    output logic y
);
    assign y = a & b;
endmodule
  
```

The port list is the pins of the box; `assign` is the gate wired inside it.

3 Testbench

Alt+A again, but this time choose **Add or create simulation sources**. File type SystemVerilog, name `and2_tb`.



The testbench has no ports. It drives stimulus into the dut and watches what comes back.

```

module and2_tb;
    logic a, b, y;

    and2 dut(.a(a), .b(b), .y(y));

    initial begin
        a = 0; b = 0; #10;
        a = 0; b = 1; #10;
        a = 1; b = 0; #10;
        a = 1; b = 1; #10;
        $finish;
    end
endmodule
  
```

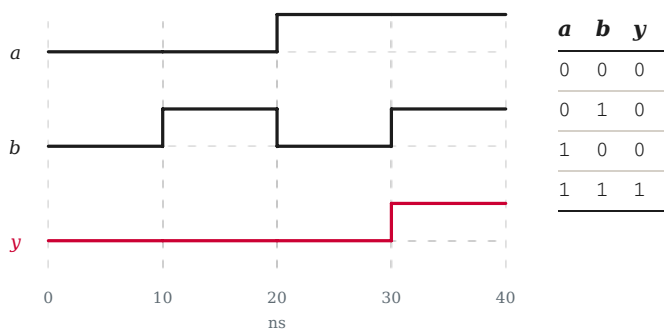
Line	What it does
<code>and2 dut(...)</code>	Instantiation — drops your design in. <code>dut</code> is the usual name: device under test
<code>.a(a)</code>	Named connection: port name after the dot, testbench signal in brackets. Always connect by name, never by position
<code>initial begin</code>	The only place that runs top to bottom
<code>#10</code>	Wait 10 time units. Simulation only — never put this in a design module
<code>\$finish</code>	Ends the run. Without it the simulation never stops

Two file sets — do not mix them

Set	File	Why
Design Sources	<code>and2.sv</code>	The hardware. This is what gets synthesised and put on a board later.
Simulation Sources	<code>and2_tb.sv</code>	Runs only in simulation. Its <code>initial</code> and <code>#10</code> are not hardware.

4 Run the simulation and read the waveform

Flow Navigator → SIMULATION → **Run Simulation** → **Run Behavioral Simulation**. The first run compiles for a few seconds, then the simulation window opens.



Try it

1. Change `&` to `|`, save, relaunch. Watch `y` become 0 1 1 1. You did not edit a line of code — you swapped an AND gate for an OR gate.